DESIGN PROCESS 08

设计进展

This chapter introduces the design and development process of Calmpa. Based on preliminary research, user experience design, and brand identity, this chapter explains how the concept of "non-intrusive rest reminders" was transformed into a physical desktop device. This chapter covers key design principles, form factor exploration, product structure, and the refinement process, demonstrating how Calmpa gradually evolved from an interaction concept into a more concrete product direction.

8.1 Design Principles

Based on preliminary research and user experience design, Calmpa adheres to four key design principles: **non-intrusive interaction**, **gentle physical feedback**, **privacy awareness**, and **desktop integration**. These principles guide subsequent form **exploration**, **structure development**, and **prototype optimization**.

① Non-intrusive Interaction



Calmpa avoids using **alarms**, **pop-ups**, **vibrations**, and **loud notifications**. Its reminders are designed to hover on the **edge of attention**, making users aware of their sedentary time without **interrupting their focused work.**

② Calm Physical Feedback



Calmpa doesn't use screen information; instead, it transforms sedentary time into **slow, petal-like movements**. This **gradual physical change** makes the sedentary state **visible** in a **calm and gentle** way.

③ Privacy-friendly Sensing



The device detects whether a user is at their desk solely through **proximity sensing**. It does **not collect facial data, emotions, work efficiency, or identity information**, thus **reducing the feeling of being monitored**.

④ Desktop Integration



Calmpa is designed to **blend into everyday office environments**. Its **compact size, stable form**, and visually appealing design that aligns with office habits make it a **part of the workspace**, rather than an added burden.

These principles guide Calmpa's **form**, **structure**, and **prototype refinement**.

8.2 Form Exploration

In terms of form exploration, the concept of **prolonged sitting time** is transformed into **petal-based body movements**, and sitting posture is integrated into **compact desktop devices**.

Motion Exploration

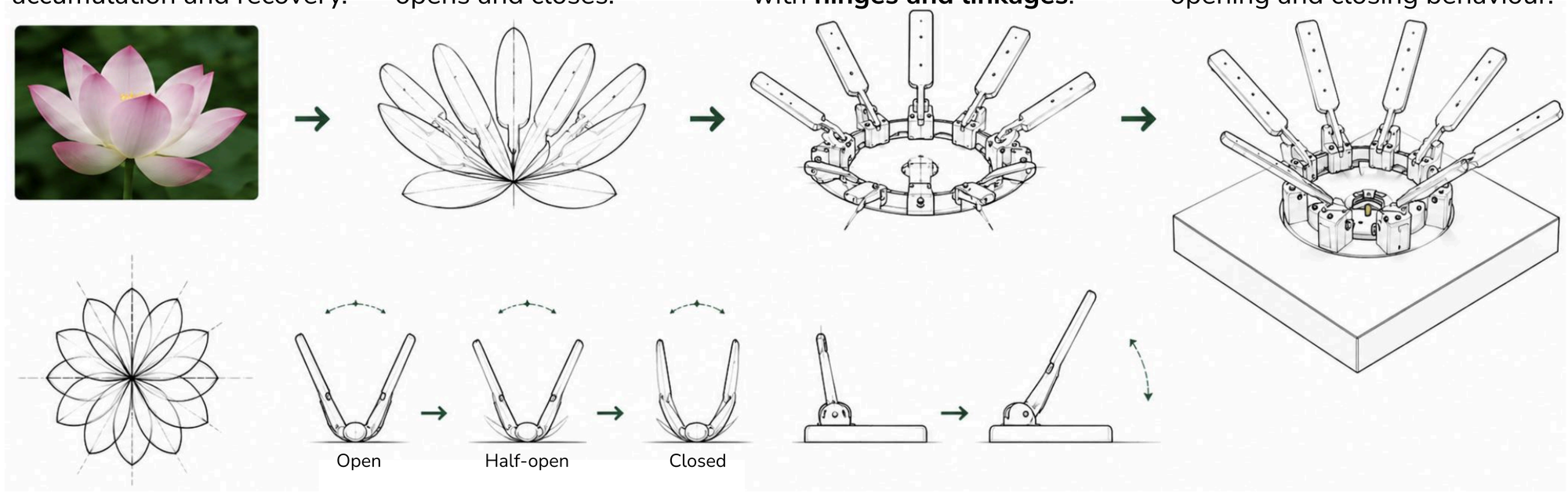
From natural inspiration to mechanical expression.

The **opening and closing** of flower petals suggests accumulation and recovery.

Abstracted into a **radial movement system** that opens and closes.

Organic curves are simplified into **repeatable linear elements** with **hinges and linkages**.

Radial arm mechanism becomes a mechanical structure about opening and closing behaviour.



Structure Exploration

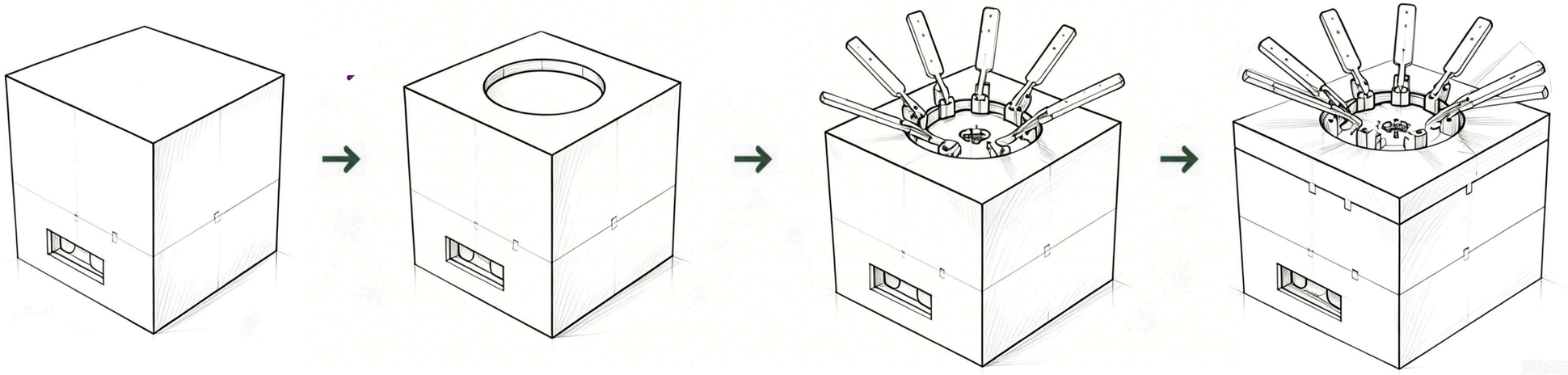
From volume to final prototype.

Start with a simple **cubic volume** to fit into desktop environments.

A **circular opening** is introduced on the top surface.

The **mechanical arms** are integrated into the circular mechanism.

The housing is **layered** for structure and a clean visual language.



8.3 Product Structure

The Calmpa structure combines **sensing**, **motion**, and **physical feedback** through a **cubic shell**, a **circular top opening**, **mechanical petal arms**, and a **front-mounted sensing module**.

Mechanical Petal Arms

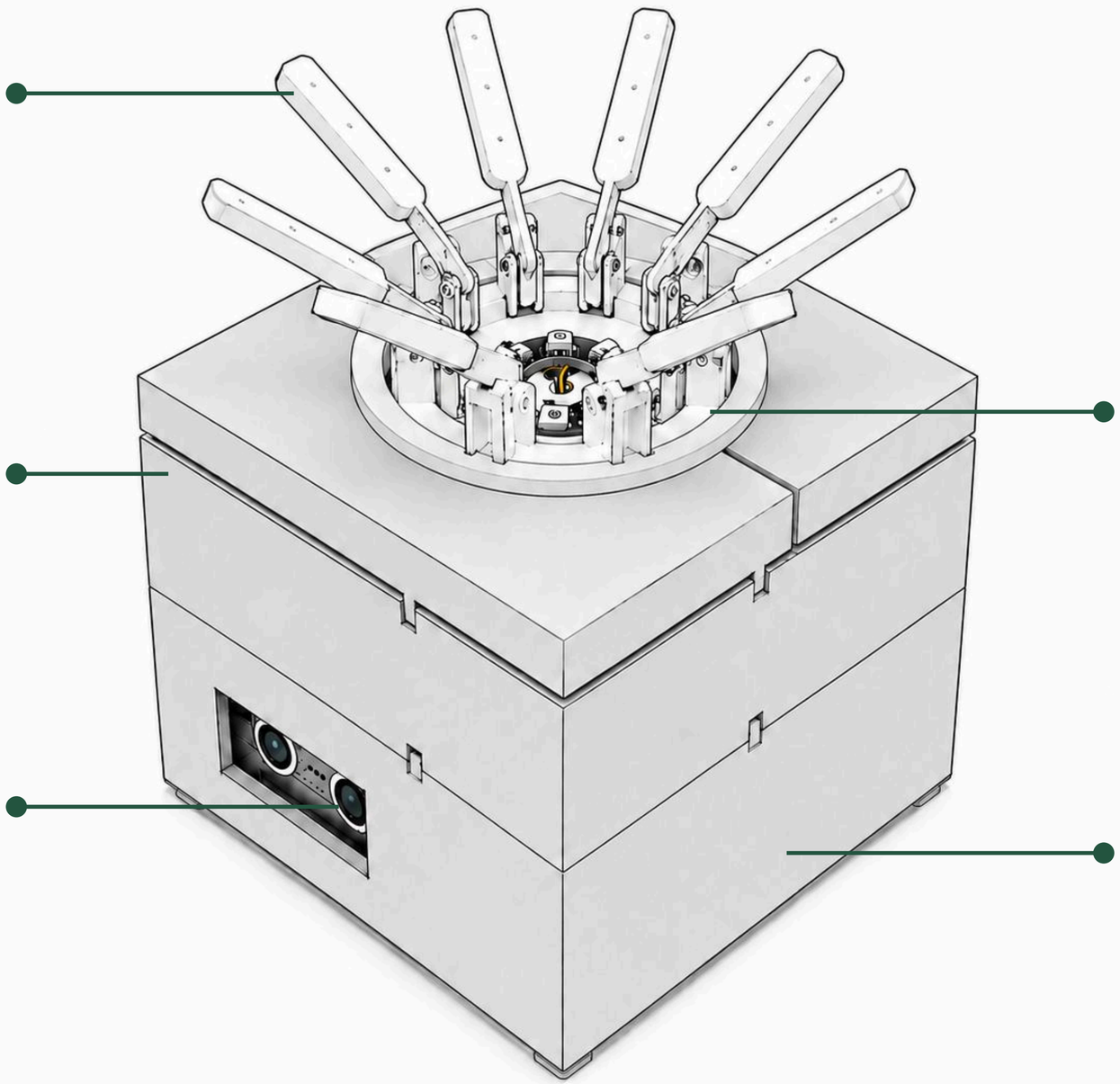
The **radial arms**, connected by **hinges and links**, open and close synchronously, generating the **primary physical feedback**.

Cubic Housing

The compact cubic shell provides **stability** and protects the **internal components**.

Front Sensing Module

The front-mounted sensing module uses an **ultrasonic sensor** to detect whether there is someone on the table.



Circular Top Opening

The circular opening integrates a r**adial mechanism**, supporting smooth **opening and closing movements**.

Layered Structure / Internal Components

The housing has a multi-layered structure to accommodate **mechanical devices**, **electronic components**, and **power management devices**.

System Flow **Desk Presence** → **Sensing** → **Control** → **Movement** → **Feedback**

8.4 Product Refinement

The improvement process focused on making Calmpa more suitable for **everyday office use**. The prototype was optimized by adjusting the **scale**, **integrating sensors**, **simplifying the visuals**, and adding more **office-friendly details**.

Scale Control

① The device's **size** has been carefully controlled to ensure that Calmpa remains within the user's **field of vision**, rather than taking up most of the desktop space. This more **compact size** helps the product remain unobtrusive when the user is focused on their work.

Sensor Integration

② The **sensing area** is integrated into the **front surface** of the housing. This reduces exposed **electronic components**, making the device look more like a **desktop product** than a **monitoring tool**.

Visual Simplification

③ The overall design has been simplified, reducing **unnecessary ornamentation**. The **clean, cubic shell** and radial mechanical structure enhance product recognizably while emphasizing **slow, physical movement** in the interaction.

Office-friendly Appearance

④ The **visual language** has been carefully designed to fit into the **modern office environment**. **Neutral tones**, clean geometric shapes, and subtle mechanical dynamism make Calmpa a part of the **workspace**, rather than an unwanted object on the desk.

These improvements make Calmpa more compact, integrated, and better suited for office use.

CONCLUSION

During its design and development, Calmpa was refined into a compact, integrated desktop device suitable for office environments. Its final design incorporates a cubic shell, circular openings, mechanical petal arms, and a front-mounted sensor module, aiming to provide soothing bodily feedback. This chapter will explain how the project transforms prolonged sitting into slow, petal-like movements, creating a product that is both practical and non-intrusive to the user.

PROTOTYPE REALIZATION 09

模型实现

This chapter introduces the prototype implementation of Calmpa. Based on previous design and development, the project was transformed into a physical prototype for testing desktop detection, servo-driven petal motion, and calm feedback functionality. Through hardware setup, system flow, motion testing, and documentation of the production process, this chapter demonstrates how Calmpa core interactive functions are implemented and validated in physical form.

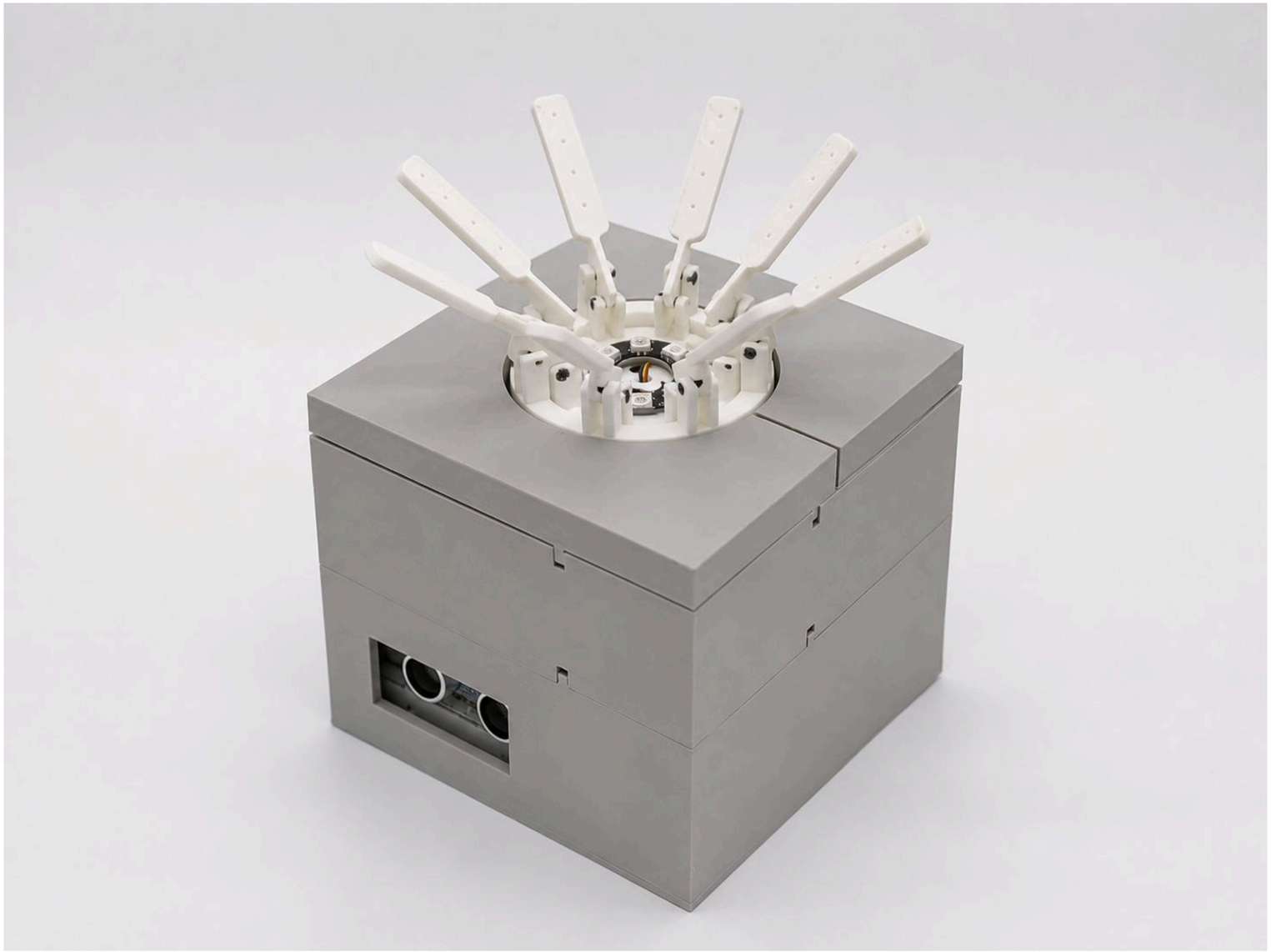
9.1 Prototype Overview

The prototype tested desktop detection, servo-driven motion, and calm body feedback.

① **Desk Detection** use an ultrasonic sensor to detect whether the user is at the desk

② **Servo-driven Movement** allow the prototype to perform opening and closing movement

③ **Calm Feedback** Prolonged sitting translates into visible bodily signals.



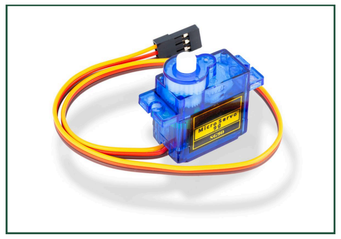
9.2 Hardware and Control Setup

This hardware device connects sensor inputs with servo-driven motion through a simple control system.

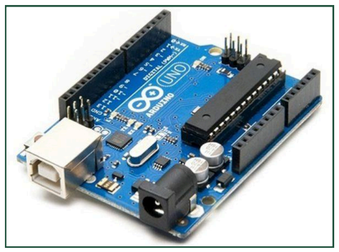
9.2.1 Hardware Setup



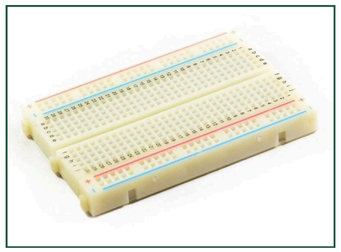
Ultrasonic Sensor
The ultrasonic sensor detects whether the user is at the desk by measuring distance.



Servo Motors
Servo motors drives the mechanical petals to achieve opening and closing movements.

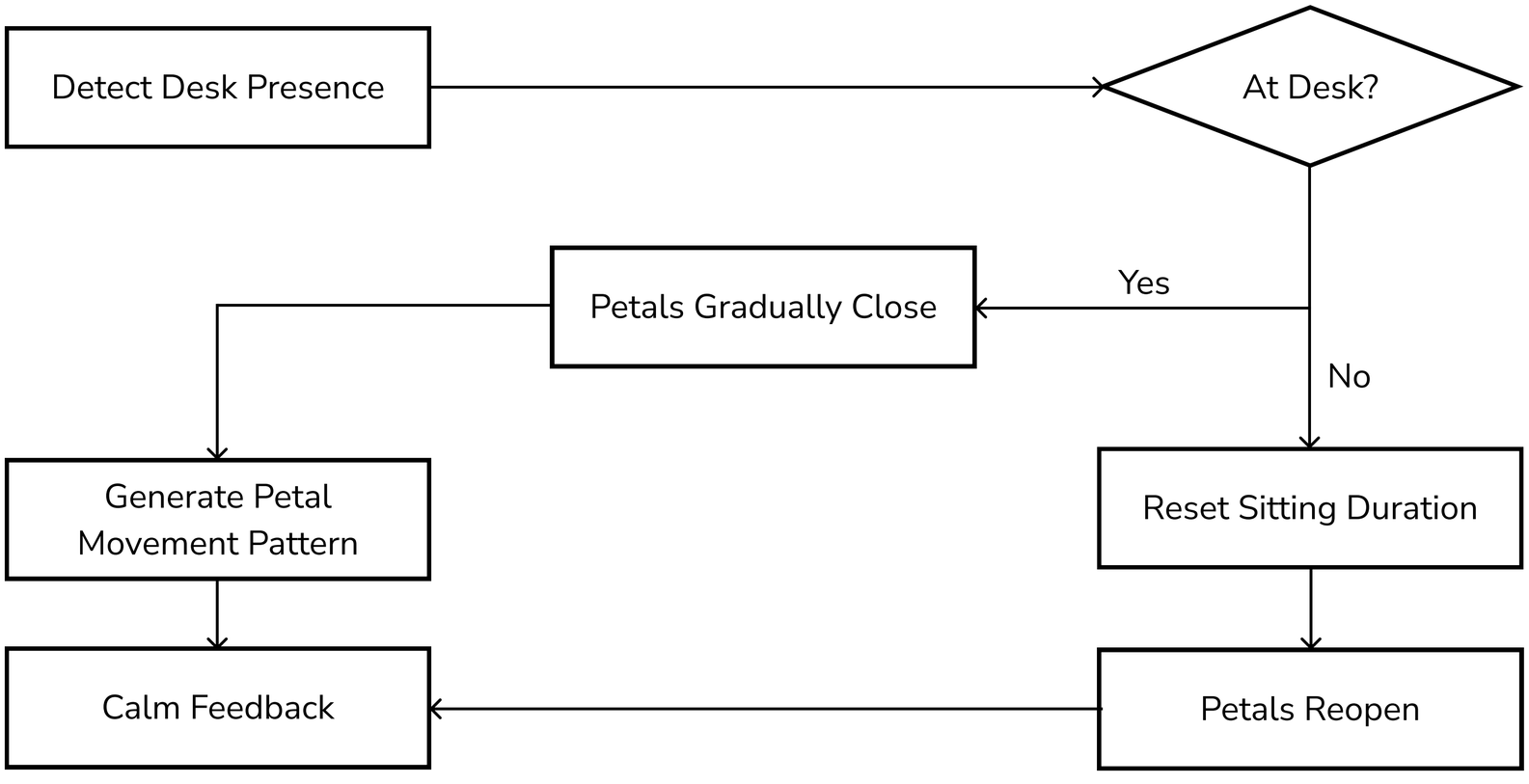


Arduino
The Arduino receives sensor data and converts it into motion commands for the petal mechanism.



Breadboard
Breadboard is used to connect sensors and servo motors without the need for soldering.

9.2.2 System Flow

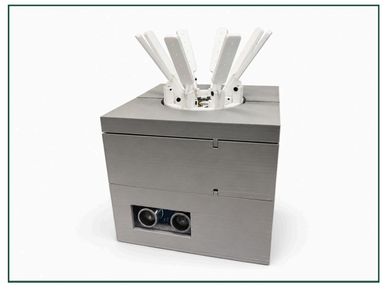


9.2.3 Motion Test

The focus of motion testing is to adjust the servo angle, motion speed, and robotic arm position to produce slow, stable, and undisturbed motion.

Open State	Closed State



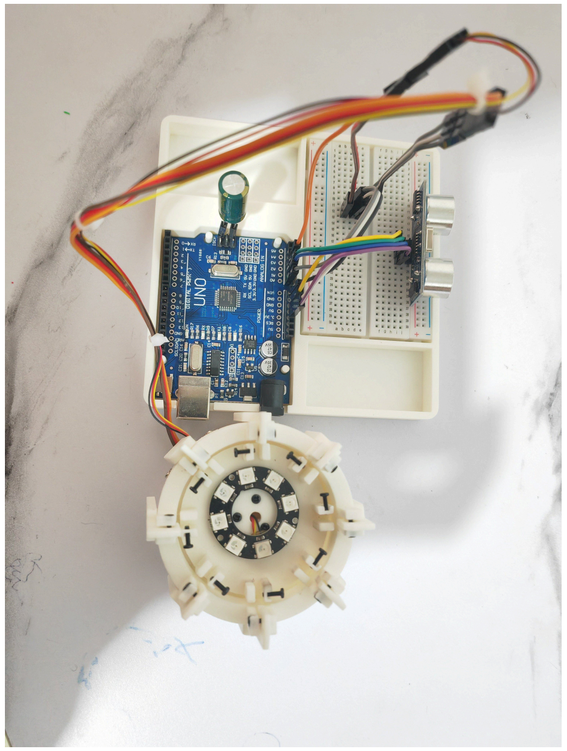


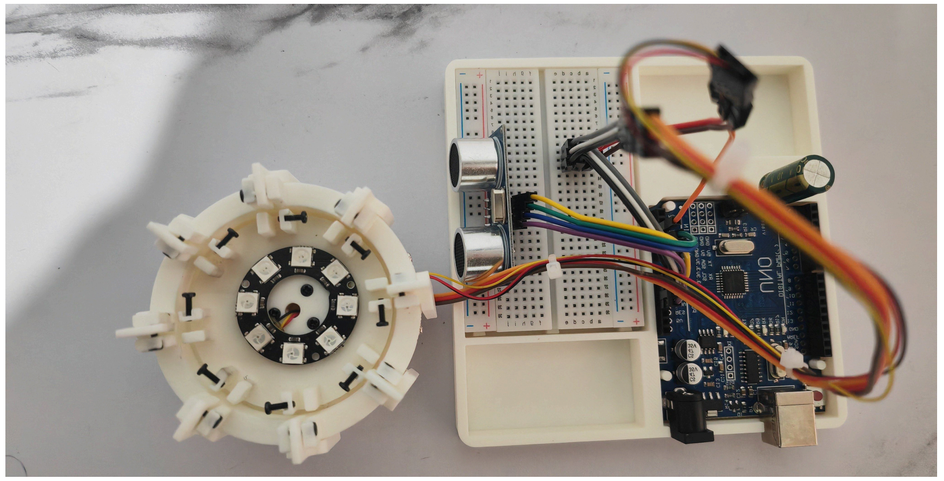
Speed Test

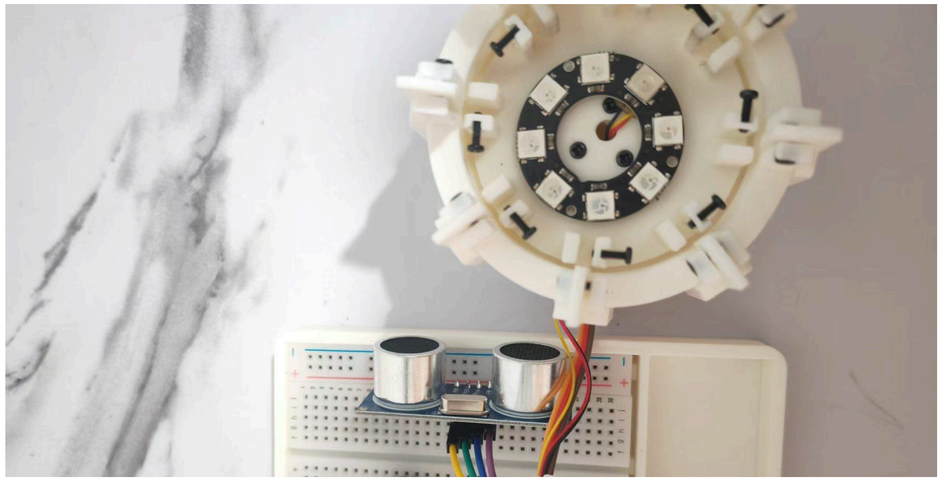
Speed Level	Movement Effect
Fast	The speed was too fast, it felt abrupt, and it was distracting.
Medium	It's acceptable, but still obvious.
Slow	**Calm, smooth and non-intrusive.**

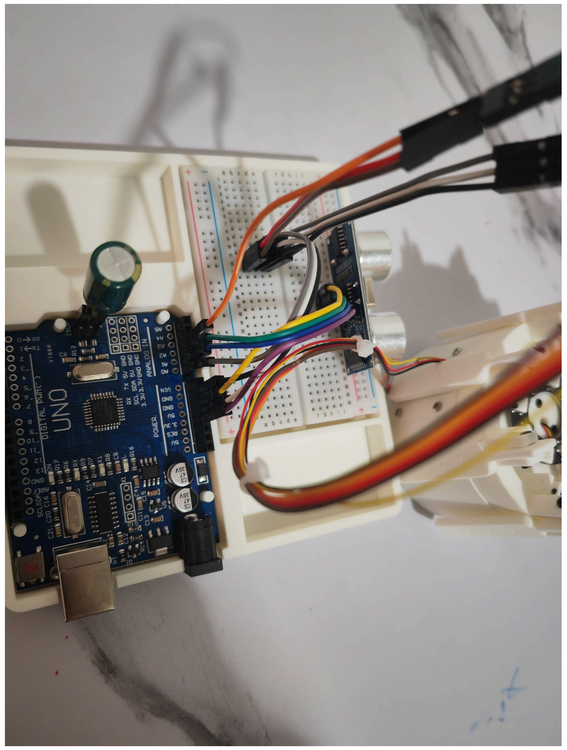
The final prototype uses **slow and gentle pace** to maintain a sense of calm on the desktop.

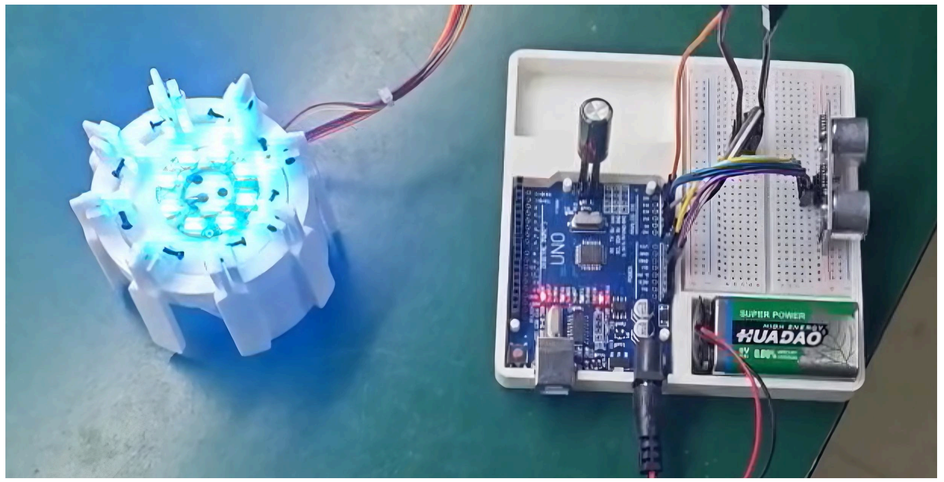
9.3 Prototype Making Process











CONCLUSION

Through prototyping, Calmpa core interaction was validated as a fully functional sensing-motion-feedback system. The prototype uses ultrasonic sensing, Arduino control, and servo-driven motion to detect the presence of a tabletop and drive the mechanical petal structure. The production process also demonstrated how the design transitioned from concept to a working prototype, laying the practical foundation for the final product.

PROTOTYPE EVALUATION 10

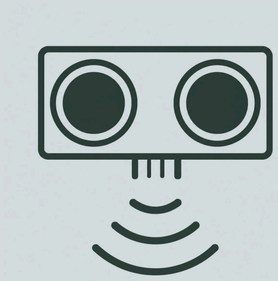
模型评估

This chapter introduces the prototype evaluation of Calmpa. Since the project is still in the prototype stage, the evaluation focuses on functional observation testing rather than informal user research. Testing includes desktop presence detection, petal motion, and motion quality, aiming to verify the feasibility of Calmpa's basic perception-motion-feedback logic.

10.1 Testing Goals and Method

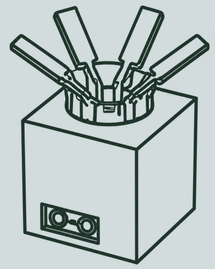
This evaluation focuses on the **basic functional performance** of the prototype, including **desktop presence detection**, **petal motion**, and **feedback mechanisms**. Since the project is still in the prototype stage, this evaluation uses **functional observation testing** rather than formal user research.

① Desk Presence Detection



Check whether the ultrasonic sensor can detect whether the user is at the desk within the actual distance.

② Petal Movement



Test whether the servo motor can drive the mechanical petal arm to switch between open and closed states.

③ Calm Feedback



Observe whether the movement feels slow, steady, and non-invasive, as a physical reminder.

Testing Method

The testing method involved placing the prototype on a table, varying the distance between the user and the sensor, and observing the movement of the petals. The testing focused on **functionality**, **motion quality**, and **interaction clarity**.

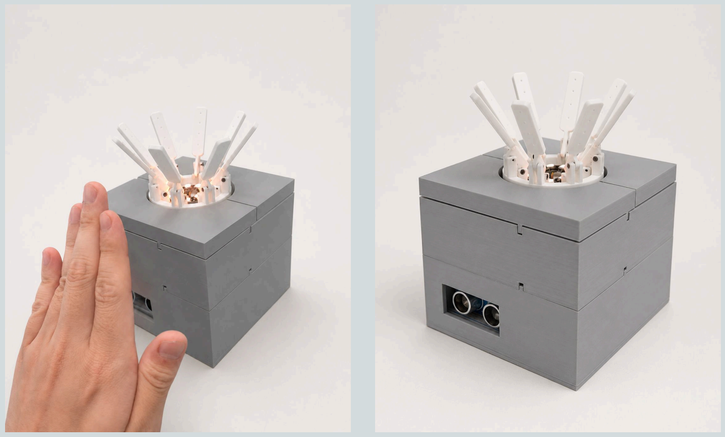
Explanation

This assessment is based on preliminary functional testing. Subsequent user testing will be conducted in the next development phase.

10.2 Functional Testing and Observation

This section documents the function test of the prototype, focusing on **sensor response**, **petal movement**, and **motion quality** during basic operation.

① Sensor Response Test



The **ultrasonic sensor** was tested at different distances. It was able to detect the presence of the table within its **actual effective range** and send a signal to the **mobile system**.

Observation

Table can be detected within a real range, but this is affected by distance and angle.

② Petal Movement Test



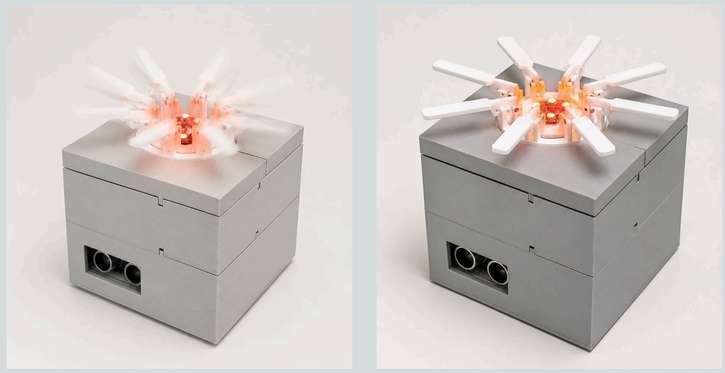
Open State | Half State | Close State

The **servo motor** successfully drove the **mechanical petal arm** to switch between **opening** and **closing states**. This confirms that prolonged sitting can be translated into **visible body movement**.

Observation

Petals can open and closes moothly. The mechanismworks as intended.

③ Motion Quality Observation



Faster Speed (More Abrupt) | Slower Speed (More Calm)

Different movement speeds were observed during the test. Faster movement felt more abrupt, while slower movement felt **calmer, more stable, and less intrusive**.

Observation

Slower speeds can provide calmer, more stable, and non-invasive bodily cues.

Summary

Tests show that Calmpa's **basic motion feedback logic** is feasible, but **sensor stability** and **motion smoothness** still need further improvement.

Explanation

This is preliminary functional testing. Long-term user testing and more detailed performance evaluation will be conducted in the next

10.3 Evaluation Summary and Limitations

Evaluation results show that Calmpa's basic sensing motion feedback logic is feasible, but further improvement is needed in term of sensor stability, motion smoothness, and product proportions.

10.3.1 Evaluation Summary

Evaluation Item	Result	Reflection
Desk Presence Detection	✔ Basic Function Achieved	Sensors can detect objects, but **distance and angle** can affect their stability.
Petal Movement	✔ Achieved	The petals can open and close, but smoother **servo calibration** is needed.
Calm Feedback	◑ Partially achieved	Slower movements are gentler, but still require **user testing** and verification.
Office Integration	◑ Partially achieved	The prototype is suitable for desktop use, but its **size can still be reduced**.

10.3.2 Limitations

① **Sensor Stability**

Ultrasonic sensors are affected by distance, angle, and surrounding objects.

② **Motion Smoothness**

The petal motion can be achieved, but the servo calibration still needs to be smoother.

③ **Product Scale**

The current prototype is still quite large, which is not conducive to long-term desktop use.

④ **User Validation**

Further user testing is needed to evaluate the reminder experience in real-world office settings.

CONCLUSION

Evaluation results demonstrate that Calmpa's basic perception-motion-feedback logic is feasible. The prototype was able to detect the presence of someone on the table and drive the petals to switch between open and closed states, showcasing the potential to create a calming atmosphere using slow body movements. However, further improvements are needed in areas such as sensor stability, motion smoothness, product size, and long-term user validation.